

Catalytic Impact of Early Investments in Advanced Clean Energy Technologies

Kick-off

PyPSA Labs, Google

September 2026



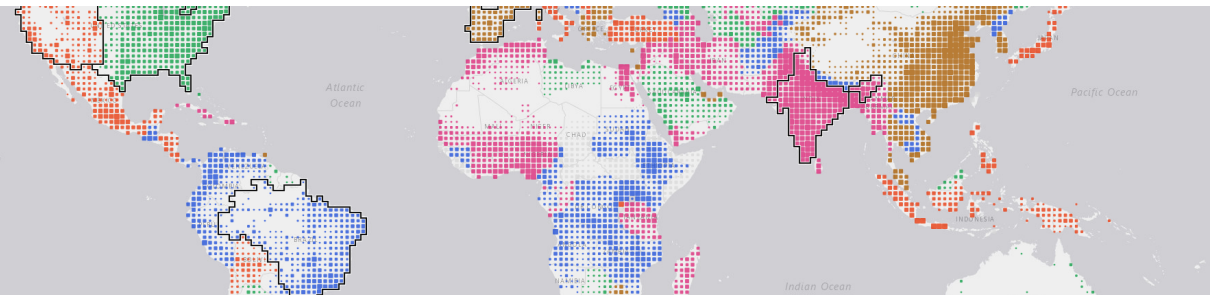
Project overview

Goal

- Quantify the catalytic impact of early investment in advanced clean technologies: experience curves, emission reductions, global addressable market

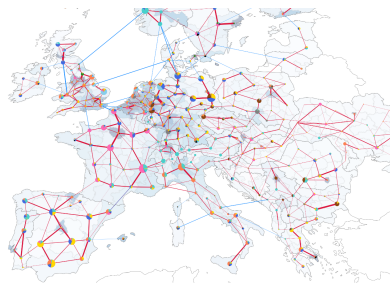
Approach

- Model representative power systems from today to mid-century in PyPSA
- Measure how early investment triggers technology learning that delivers long-term emission reductions



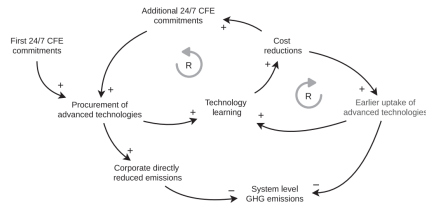
Power System Modelling

- PyPSA for capacity expansion and dispatch optimisation
- PyPSA-Earth as the global data pipeline: brownfield fleet, demand, grid topology
- atlite for renewable resource potentials and capacity factors
- Five power systems, each representing a *grid archetype*, support a qualitative extrapolation to the global addressable market

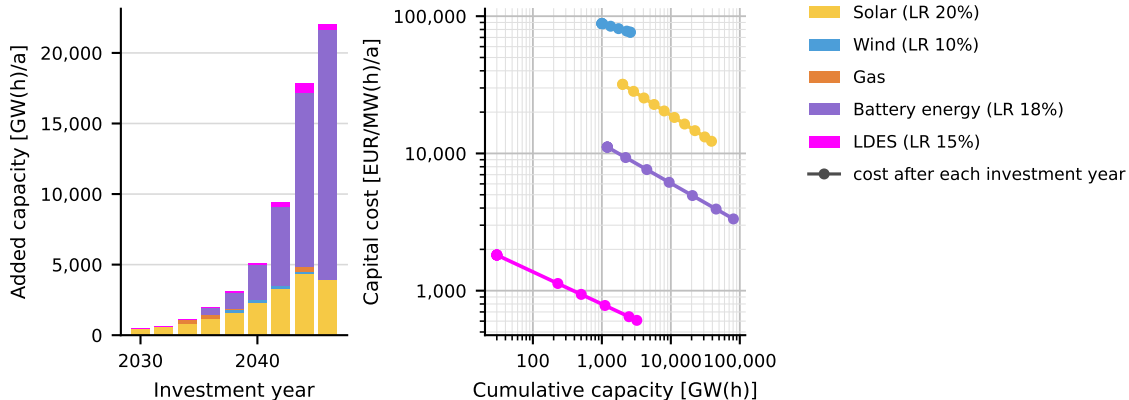


Catalytic Investment Mechanism

- Each system is modelled for 2025, 2030, ..., 2050
- 2025 serves as a basic validation year
- 2030 to 2050 are first run as a “baseline” scenario without catalytic investment
- In *catalysed* runs, advanced technologies are then pulled into the market, triggering technology learning and lower investment costs in subsequent years



Prototype: Endogenous learning in the world model



The Initial Investment Mechanism

TBD

(in decision items)

Scenarios

- **Wind/solar acceptance limits:** land scarcity (e.g. Singapore), protected areas, ...
- **CDR breakthrough:** engineered CO₂ removal becomes cheap (e.g. 150 \$/tCO₂ by 2040)
- **Advanced baseload breakthrough:** SMRs or another firm clean technology see a rapid cost decline
- **Advanced geothermal breakthrough:** some regions gain access to high-temperature bedrock through advanced drilling
- **Solid oxide fuel cell breakthrough:** cheap backup power from hydrogen or from natural gas with carbon capture

Goals and Priorities

- Results are framed to be **generally applicable to any investor**, expressed as a “procurement volume” or a “green premium volume”
- Everything is open source; our contributions are fed back into the model ecosystem
- The modelling abstracts from forward-looking national energy and climate policies, focusing on the brownfield fleet, announced retirements and an exogenous CO₂ price

Decision Items

- Technologies
- Initial investment mechanism
- Modelled regions

Technologies

Mature: exogenous costs

- Solar PV, onshore/offshore wind
- Hydro: reservoir, run-of-river, pumped hydro
- Biomass
- Nuclear (large LWR)
- Gas CCGT and OCGT
- Li-ion battery

Advanced: endogenous learning

- LDES: vanadium flow, CO₂ battery, iron-air, H₂ + SOFC
- Nuclear SMR
- Next-gen geothermal: EGS, closed-loop
- Allam-cycle gas + CCS
- Solid oxide fuel cell
- Fusion (with caution)
- CDR: direct air capture, cost cut-off

The Initial Investment Mechanism



- A well-tested mechanism for early investment that yields a general investment framing: investment volume, emission savings, etc.

Modelled regions and grid archetypes



Renewables-constrained, high-density

☐ modelled: North-West Europe



Geothermal-favourable

☐ modelled: United States (West)



High renewables potential

☐ modelled: India



Clean incumbents (hydro & nuclear)

☐ modelled: Brazil



Fossil-heavy brownfield with large economies

☐ modelled: United States (East)

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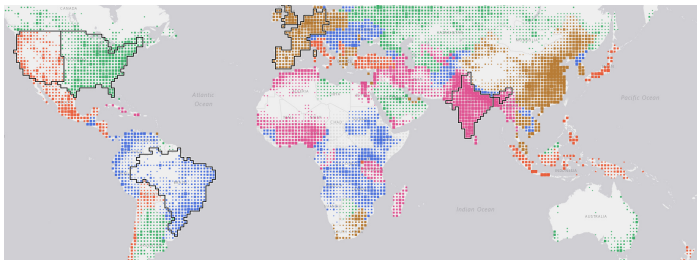
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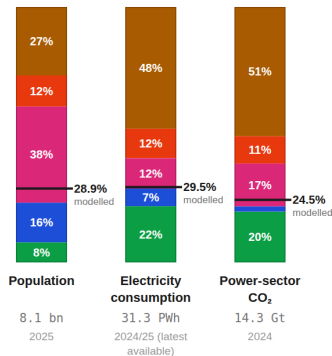
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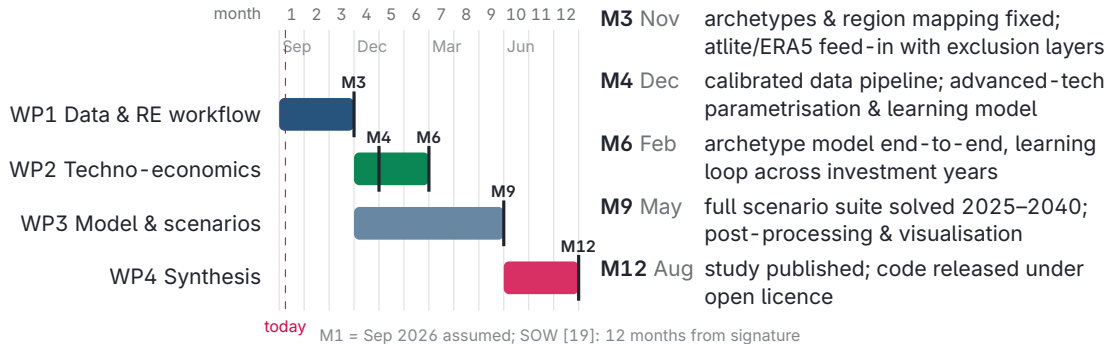


Fossil-heavy brownfield with large economies

□ modelled: United States (East)

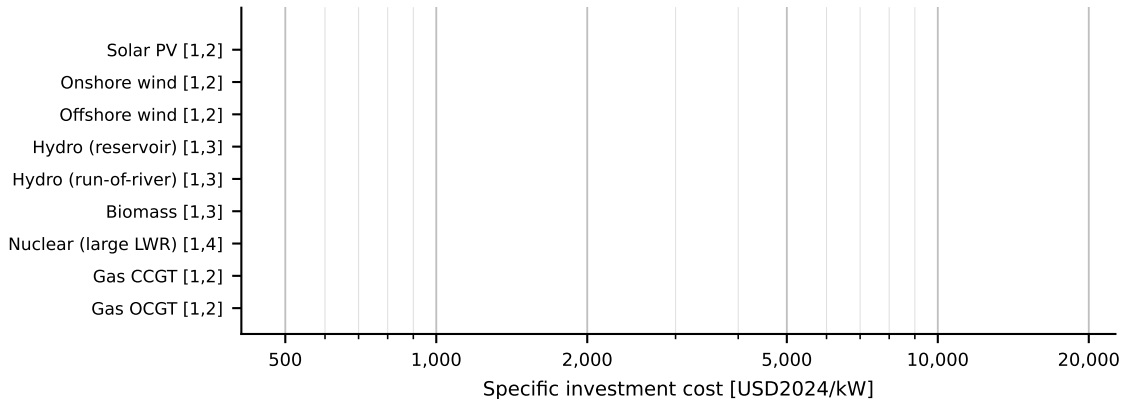


SOW work packages and milestones

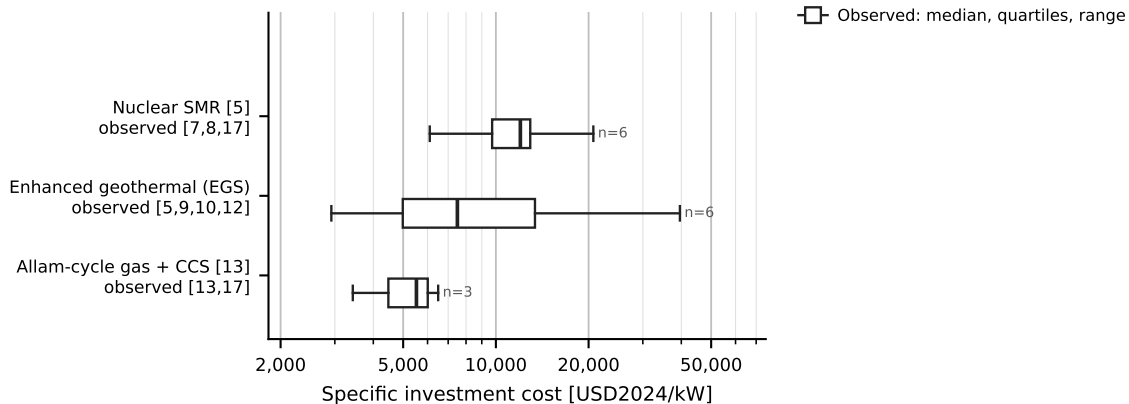


Open Discussion

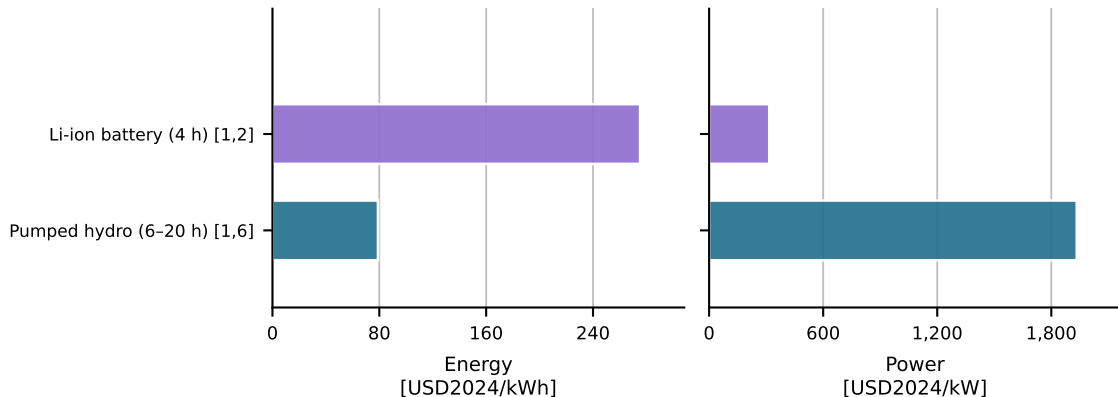
2025 capex baseline: mature generation



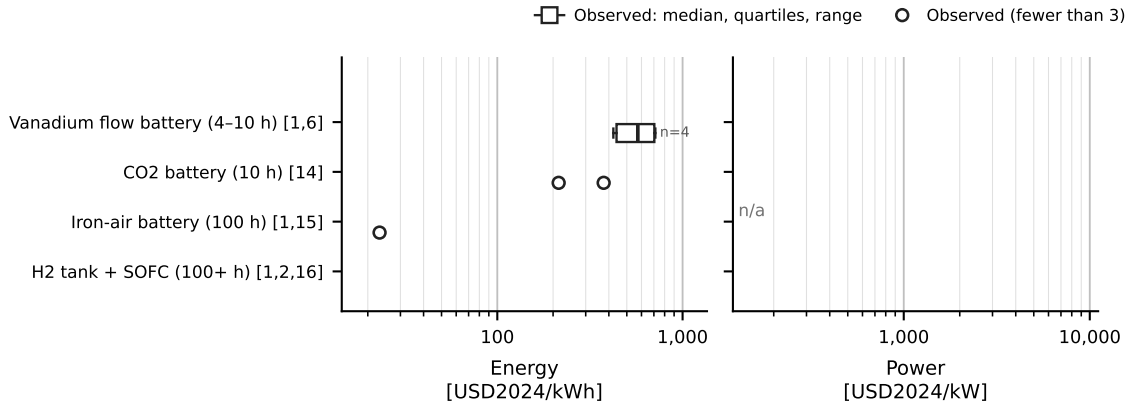
2025 capex baseline: advanced generation



2025 capex baseline: mature storage



2025 capex baseline: advanced storage



References

- [1] PyPSA, technology-data v0.15.0 (2025), cost assumptions for 2025 github.com
- [2] Danish Energy Agency, Technology Data catalogues: electricity generation; energy storage (2024/25) ens.dk
- [3] Schröder et al. (DIW Berlin), Costs of Electricity Generation until 2050, Data Documentation 68 (2013) diw.de
- [4] Lazard, Levelized Cost of Energy+ (LCOE+), June 2025 and v16.0 (2023) lazard.com
- [5] NREL, Annual Technology Baseline 2024, electricity (v3 data) atb.nrel.gov
- [6] PNNL / DOE, 2022 Grid Energy Storage Technology Cost and Performance Assessment (Viswanathan et al.) pnnl.gov
- [7] EIA / Sargent & Lundy, Capital Cost and Performance Characteristics for Utility-Scale Generating Technologies (AEO2025) eia.gov
- [8] US DOE, Pathways to Commercial Liftoff: Advanced Nuclear (Sep 2024 update) liftoff.energy.gov
- [9] US DOE, Pathways to Commercial Liftoff: Next-Generation Geothermal Power (Mar 2024) liftoff.energy.gov
- [10] NREL, Enhanced Geothermal Shot Analysis for the Geothermal Technologies Office (2023) nrel.gov
- [11] BloombergNEF, Energy Storage System Cost Survey 2024 (as reported by Energy-Storage.news) about.bnef.com
- [12] Fervo Energy, Form S-1 registration statement (2026), Cape Station cost disclosures sec.gov
- [13] NET Power, Q4 2024 results and business update (10 Mar 2025), Project Permian cost estimate ir.netpower.com
- [14] Energy Dome, CO2 Battery cost and performance statements (2022–2024) energydome.com
- [15] Form Energy, iron-air modelling recommendations (2023) and cost target statements formenergy.com
- [16] Bloom Energy, product cost disclosures 2017–2023 (via Thunder Said Energy); Energy Server datasheet (2026) bloomenergy.com
- [17] Developer, regulator and press disclosures of individual projects (Vogle, Hinkley Point C, Flamanville 3, Darlington, Kemmerer, Snowy 2.0, Fengning, Dalian, Alliant, ...); per-project sources in the repository github.com
- [18] T. Brown, priam-myopic: myopic world electricity model with endogenous learning (prototype, 2026) github.com
- [19] PyPSA Labs and Google, Statement of Work: Catalytic Impact of Early Investments in Advanced Clean Energy Tech (2026)

Frame title

DEMONSTRATION

Text with **alerted text**, the accents green, red, gray and chips that highlight quietly or loudly [1].

- First item of an itemisation
 - Second item with **bold** text
 - Third item with a [hyperlink](#)
-
- 1 First item of an enumeration
 - 2 Second item of an enumeration

Block

Uses the primary brand colour.

Example block

Uses the green accent.

Alert block

Uses the red accent.

Consider a total system cost minimisation problem¹:

$$\min_{g_{s,t}, G_s} \quad \overbrace{\sum_s c_s G_s}^{\text{investment costs}} + \overbrace{\sum_{s,t} o_s g_{s,t}}^{\text{fuel costs}} \quad \text{s.t.}$$

$$\sum_s g_{s,t} - d_t = 0 \quad \perp \quad \lambda_t \quad \text{(nodal energy balance)}$$

The dual variable λ_t of the energy balance is the **market price**.

¹Assume the investment costs are annualised.

A statement slide for one idea.

1. Brown, T., Hörsch, J. & Schlachtberger, D. PyPSA: Python for Power System Analysis. *Journal of Open Research Software* 6. doi:[10.5334/jors.188](https://doi.org/10.5334/jors.188) (2018).

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REFERENCES



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